

E-commerce Mid Exam —

PDF 1 Topics

1. Definition of E-commerce

E-commerce means using the Internet and Web to do business. In simple words, it is buying and selling that happens online between people or companies. Any time money or value moves from one side to another through the internet, that is e-commerce. Examples are placing an order, paying online, or sending money digitally. It is different from e-business because e-commerce always happens between two different parties, like a buyer and a seller. Amazon, Daraz, and bKash are common examples where digital buying and selling replace old face-to-face shopping.

2. E-business with Example

E-business means using digital technology, mainly the internet, to run business work like buying, selling, marketing, and customer service. It is about using digital systems inside one company only. It does not need to cross between two different companies like e-commerce does. Example: a company using software to manage its own staff salary or stock is doing e-business. Netflix using digital systems to manage and deliver its shows is also an example. So e-business is more about internal digital work, not always buying/selling with outside people.

3. Mass Marketing

Mass marketing means a business sends the same ad or message to everyone, without thinking about what each person likes. This was common in old-style traditional commerce. For example, a newspaper car ad is seen by thousands of people, but only a few of them may actually want a car. Still, the same ad goes to all readers equally. Nobody's personal choice is considered here. E-commerce changed this by allowing personalized ads, where each customer can see different offers based on their own interest and past searches.

4. Information Asymmetry

Information asymmetry means the seller knows more about a product than the buyer does. The buyer has to trust the seller's words because they cannot check everything themselves. This happens a lot in traditional commerce. Example: in the 1980s, when buying a used car, the buyer may not know if it had an accident before, but the seller already knows this. The buyer has very little way to check unless they bring an expert with them. In e-commerce, this problem becomes smaller because of online reviews, ratings, and open information for everyone to see.

5. Unique Features of E-commerce Technology

E-commerce technology has eight special features that make it different from old commerce:

1. Ubiquity – available everywhere, all the time
2. Global reach – can reach people in any country
3. Universal standards – same technology rules work for everyone
4. Information richness – can show video, audio, and text together
5. Interactivity – buyer and seller can talk two-way, not just one-way
6. Information density – information becomes cheaper, easier, more correct
7. Personalization/Customization – message can be made specially for one person
8. Social technology – users can create and share content easily

These features together make online shopping faster, cheaper, and more fun than shopping in a normal store.

PDF 2 Topics

6. Message Distribution Infrastructure

This layer is software that sends and receives messages between a server and a client (user). Its job is simply to carry the message safely from one side to the other, like sending an HTML page from a website to your browser. Messages can be plain, like a normal email, or in a fixed format, like an order form. Common tools in this layer are EDI, e-mail, and HTTP. Example: when a student sends an email to a teacher through Gmail, the message goes from student's Gmail app to Gmail's server using SMTP rule, and then reaches the teacher's inbox. This whole sending process is message distribution infrastructure.

7. Business Services Infrastructure

This layer's main job is to support normal business work, especially collecting payment safely. It protects credit card numbers using encryption (a way to hide information) and helps transfer money electronically (EFT). It also checks if the customer is real, which is called authentication. Example: when someone buys a laptop from Amazon with a card, the payment system hides the card details, the bank checks the buyer using OTP, and then money moves from buyer's account to seller's account. This layer makes online payment safe and trustworthy for everyone.

8. E-commerce Application

An e-commerce application is the final thing a customer actually sees and uses, built on top of all other layers. Example: think of an online book shop. The application itself is the book list (catalog), encryption keeps the credit card number safe, the page is built using HTML, HTTP carries the messages, and the Internet moves everything between buyer and seller. So every layer below — network, message sending, web publishing, and business service — works together just to make this one simple book shop app work properly.

9. Difference between PDF and HTML

HTML is a markup language, while PDF is a page description language. HTML can be made using many programs, even simple word processors, and can be opened free using any browser. PDF needs special software from Adobe to create, though the reader/viewer is free for everyone. HTML only saves the structure of a page, but PDF saves both structure and exact look/layout. Both HTML and PDF files can link to each other. The main difference: in HTML, the reader can change how the page looks using their own browser settings, but in PDF, only the creator decides the final look, and nobody can change it.

PDF 3 Topics

10. Business Model and Business Plan

A business model is a plan of activities made to earn profit in the market. It is the center part of a business plan. A business plan is the full written document that explains this business model in detail. An e-commerce business model specially uses the power of Internet, Web, and mobile to make profit. Simply put, business model tells "how the company will earn money," and business plan is the paper/document that shows this model to investors or other people.

11. Eight Key Elements of Business Model

A strong business model must have these eight parts:

9. Value proposition – why customer should buy from you
10. Revenue model – how company will earn money
11. Market opportunity – which market the company wants to serve
12. Competitive environment – who else is selling in same market
13. Competitive advantage – special benefit the firm has over others
14. Market strategy – how to promote and attract customers
15. Organizational development – how the company organizes its work
16. Management team – experience and skill of company leaders

Every part is important, and missing even one part makes the whole model weak.

12. Value Proposition

Value proposition means how a product or service solves the customer's need. In e-commerce, main value points are personalization, convenience, low price or free shipping, fast delivery, big selection, and good quality. Example: on Amazon, when Sarah searches for headphones, she gets personal suggestions, can compare many options easily, may get free shipping with Prime, and receive fast delivery. All these points together give customers a strong reason to choose one shopping site over another.

13. Revenue Model

Revenue model explains how a company plans to make money. There are five main types: advertising (earning from ads, like Facebook), subscription (fixed fee for access, like Netflix), transaction fee (small charge per transaction, like eBay), sales (direct selling of goods, like Amazon), and affiliate (earning through referral, like hotel booking sites). Many companies use more than one revenue model together to earn steady income from different sources.

14. Competitive Advantage

A firm gets competitive advantage when it makes a better product or sells it cheaper than most competitors. Important ideas here: asymmetry means one company has more resources than others; first-mover advantage means being first in the market gives extra benefit. Unfair advantage comes from things hard to copy, like a strong brand name or big investment. Leverage means using big existing resources, like a large customer base, to grow into new markets. In a perfect market, competitive advantage does not exist because everyone has equal access to resources.

15. B2C Models: E-tailer, Transaction Broker, Service Provider

E-tailer: Online version of a normal shop, like Walmart online. Earns money through direct sales. Types include virtual merchant (online only), bricks-and-clicks (both physical and online store), catalog merchant (sells through online/printed catalog), and manufacturer-direct (brand sells directly, like Nike's own site).

Transaction Broker

A company that helps people finish online deals but does not own the product itself, like Hotels.com. Its main value is saving time and money for customers. It earns money through transaction fees.

Service Provider

A business that gives online services instead of physical goods, like Gmail or PayPal. Its value is convenience, saving time, and low cost. It earns through service sales, subscription, or ads.

16. Definitions: Seed Capital, Strategic Partner, Angel Investors, MRO

- Seed capital: The founder's own personal money used to start a business at the very beginning.
- Strategic partner: A company that helps another business grow, not just with money but also with tech, skill, or customer network.
- Angel investors: Rich individual people who put their own money into a business in exchange for a share of ownership, like in Shark Tank.
- MRO (Maintenance, Repair, Operations): Items used during production that don't become part of the final product, such as cleaning tools, repair parts, and office supplies.

17. B2B Models: E-distributor, Exchange

E-distributor: Like an online wholesale/retail store for MRO and indirect goods, owned by one company that serves many business buyers. Offers a big online catalog for search, compare, and bulk order. Earns through direct sales. Example: Grainger.com.

Exchange: A separately owned online marketplace where many suppliers meet a few big buyers. Creates strong price competition among sellers. Earns through transaction/commission fees. Over time, many exchanges closed because of this heavy price competition.

Class Test Style Questions

Explain the major revenue models

There are five main revenue models used in e-commerce:

17. Advertising model – earns money by showing ads (example: Facebook)
18. Subscription model – earns a fixed fee for giving access to content (example: Netflix)
19. Transaction fee model – earns a small charge for each completed deal (example: eBay)
20. Sales model – earns by selling goods or services directly (example: Amazon)
21. Affiliate model – earns money through referral fees for sending customers to another business (example: hotel/airline booking sites)

Many businesses use more than one model together to earn steady income.

What is a transaction broker? Explain its primary value proposition and revenue model

A transaction broker is a company that helps customers finish online deals without owning the product or service itself. It works as a middleman between buyer and seller, making the process fast, easy, and safe.

- Primary value proposition: Saves time and money for customers by making online deals simple, such as trading stocks or making card payments.
- Revenue model: Earns through transaction fees, meaning it gets a small charge each time someone completes a deal through the platform.

Mention three industries where the transaction broker model is commonly used

22. Financial services – like E*TRADE, where people trade stocks and manage money online
23. Travel services – like Hotels.com and Expedia, where people book hotels or flights without the platform owning them
24. Job placement services – like LinkedIn Jobs and Indeed, where companies and job seekers connect, and the platform earns fees for that connection